

characteristics of the solar panel power output, devise technical maximum power point tracking (MPPT) algorithms to maximize the energy delivery, and understand complex power stage design and control. The kit, priced at US\$ 850, introduces designers to a fully suitable MCU and methodically addresses the application challenges by breaking down the development process into manageable pieces.

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Pick-and-place system

Essemtec has upgraded its Pantera X-plus pick-and-place system for small batches, incorporating new features for



highly-flexible small batch assembly. The enhanced X-vision system has improved image processing for small chip components and com-

plements the existing on-the-fly laser centring feature. The combination of fast laser centring and high-precision image processing now allows customers to achieve the best possible placement results in this machine segment.

By using a new digital image top camera, image recording of fiducials can be optimised, thereby improving the detection of the PCB alignment. The Pantera X-plus is no longer compatible with Microsoft's Windows XP operating system, but works with the latest Windows 8 version.

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TEST & MEASUREMENT

Portable oscilloscopes

Agilent Technologies Inc. has introduced new high-performance portable oscilloscope series deploying next-generation oscilloscope technology. The Infiniium S-Series has the world's

fastest 10-bit ADC. Compared with traditional scopes with 8-bit ADCs, the S-Series has four times the vertical resolution for precise viewing of signal detail.

Coupled with a new low-noise front end, the S-Series delivers an ENOB (effective number of bits) for the system of more than eight, the highest in the industry. The series includes bandwidths from 500MHz to 8GHz with four-channel DSO models and 16-digital-channel MSO models. Standard memory is 50Mpts per channel or 100Mpts per channel when interleaved, the deepest in the industry.

The S-Series features application-specific measurement software, fuelled by a powerful motherboard with 8GB RAM. The 38.1cm (15-inch) multi-touch capacitive display is the largest in the industry, making it easier to see results and easier to use.

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LEDs & LED LIGHTING

Mid-LED package

Everlight Electronics has introduced the ultra-thin 940nm top-view infrared LED



IR92-01C/L491/2R in a miniature mid-LED package. The new 940nm IR LED comes in a small 2mm×1.4mm package with an ultra-low thickness of 0.7mm. This makes it ideal for all kinds of space-constrained end products like tablets and smartphones.

Along with a low forward voltage of 1.3V at 20mA, a high output power of 25mW/sr at 70mA, and a narrow viewing angle of 45° for even higher output intensity without lens, the IR92-01C series is perfectly suited for IR data transmission applications.

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Chip-on-board arrays

Philips Lumileds has launched chip-on-board (CoB) arrays for parabolic aluminium reflectors (PAR) 38 equiv-



alent lamps, which achieve 10 per cent or greater efficacy than competing solutions, according to the company. Also

ideal for spotlights, the Luxeon CoB 1202 has a typical efficacy of 115lm/W, which varies from 95-130lm/W depending on the colour temperature and CRI of the array. It is available in a single 3-step as well as a single 5-step MacAdam Ellipse, ensuring uniform optical performance in the application.

Ideal applications include downlights and directional lamps. The arrays are all hot-tested at 85°C, which means that the luminaire design is simplified and testing can be minimised.

Philips Lumileds India distributor: Key Operation and Electromechanics Pvt Ltd, New Delhi
Phone: 8377806486

Website: www.philipslumileds.com

POWER SUPPLIES

Tubular batteries

Base Batteries has launched Base Tuff-BT 500 tall tubular batteries for home UPS application. These are the energy-efficient backup units designed to protect homes from experiencing the inconvenience caused by power outages, minimise energy consumption and help the consumer save on electricity bills. The tall container allows for greater amount of electrolyte, which reduces maintenance and increases life even in tough power conditions.



BT series is environment-friendly and free from acid fumes, indicates water level and has a special tubular positive-plate design for long life in deep-discharge cycle.

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